

Technology Stack

Date	15 July 2026
Team ID	TM-RISINGWATERS-04
Project Name	Rising Waters: AI Powered Flood Prediction System
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side (e.g., React, Angular, HTML/CSS)	React (TypeScript) & Tailwind CSS	React provides a component-based model ideal for highly modular dashboards. TypeScript enforces strict types, reducing runtime errors, and Tailwind CSS provides fast, clean responsive UI styling for crisis coordinators. Includes <i>Leaflet.js</i> for real-time map spatial rendering.
2	Backend / Server-Side (e.g., Node.js, Python/Django, Java)	Python (FastAPI) & Uvicorn	FastAPI is an asynchronous, high-performance web framework. Python is native to modern machine learning workflows, facilitating seamless ingestion and execution of our custom neural network inundation risk models.
3	Database / Data Storage (e.g., MySQL, MongoDB, PostgreSQL)	PostgreSQL (with PostGIS) & Redis	PostgreSQL handles highly structured user configurations and historic event logs. The PostGIS extension natively optimizes geographic query tasks. Redis acts as an in-memory database caching real-time active system warnings for speed.
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Heroku, Docker)	Docker & Kubernetes (AWS EKS)	Containerization guarantees development environments match staging/production. Deployment on Kubernetes allows dynamic horizontal auto-scaling of API server replicas during extreme storm situations when traffic surges.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub & GitHub Actions	GitHub supports structured trunk-based peer development. GitHub Actions automates visual pipeline tests, linting scans, build compilations,

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
			and auto-pushes production-ready images directly to Docker registries.
6	Third-Party APIs / Other Tools <i>(e.g., Stripe, Twilio, Google Maps)</i>	TensorFlow, Twilio API, & OpenWeatherMap API	TensorFlow structures the core predictive deep learning workflows. Twilio delivers instantaneous emergency SMS alerts to targeted regions, and regional weather APIs provide live forecast parameters.